DO NOW — RECALL

Software testing: finding and correcting faults

Name: _____ Class: _____ Date: _____

1. Define a cache miss.

2. When is A AND B true?

3. Give a boundary-invalid input just below 1.

4. Does 61 satisfy `seconds>=1 || seconds<=60`?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Requested result is not already stored.

2. Model answer

Only when both A and B are true.

3. Model answer

0.

4. Model answer

Yes, because seconds>=1 is true.

Vocabulary — know these terms

logic error — Runs but produces the wrong result.

inclusive boundary — Endpoint allowed by a range.

validation — Check input meets constraints.

debugging — Find and correct a fault.

regression test — Repeat tests after a change.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

AND requires both limits to hold and correctly rejects 61.

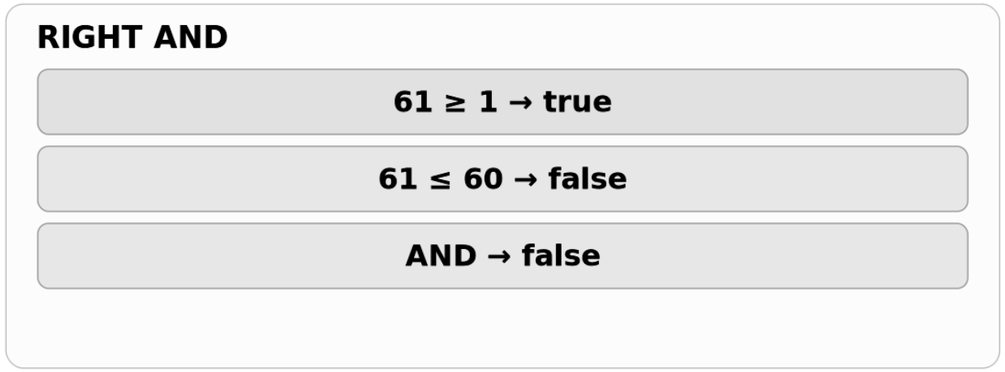
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A duration validator for 1–60 accepts 61 because it uses OR. Correct it.

OR accepts 61 because only one side needs to be true.



RIGHT AND; 61 ≥ 1 → true; 61 ≤ 60 → false; AND → false

NEW LEARNING

Read, annotate and explain

Read the stated rule

Because the permitted ages are inclusive, both age 12 and age 18 must be accepted.

Locate the Boolean error

The strict comparison operators > and < exclude both valid boundaries.

Repair and test

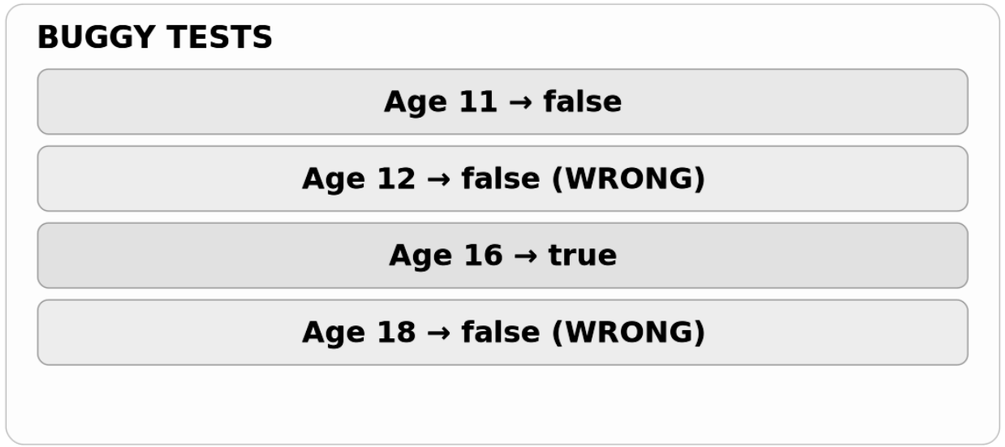
Change to >= and <=; retest the boundaries plus values just outside the range.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

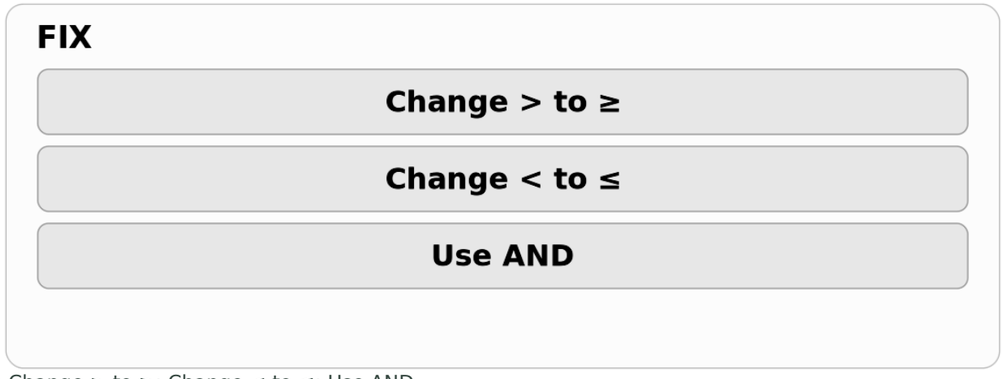
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BUGGY TESTS; Age 11 → false; Age 12 → false (WRONG); Age 16 → true; Age 18 → false (WRONG)

New Learning



FIX; Change > to ≥; Change < to ≤; Use AND

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

For a bounded range, OR usually replaces AND without changing the outcome.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

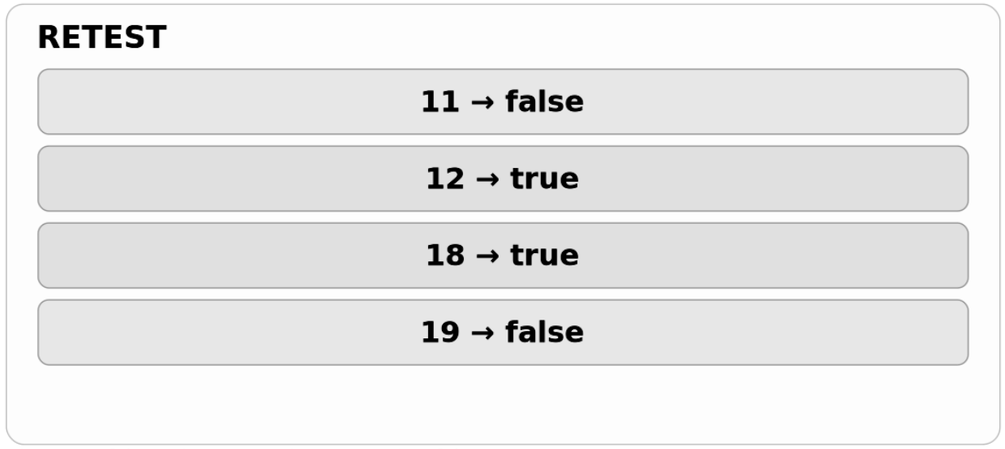
Question 3

Outline a debugging and regression-testing process after correcting a validator.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



RETEST; 11 → false; 12 → true; 18 → true; 19 → false

I DO — WORKED EXAMPLE

Follow the worked method

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

Predict first

The valid endpoints were incorrectly rejected.

Model and check

Change the operators to >= and <=, and test −1 and 11 as invalid values.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Correct x>5 && x<15 for values 5–15 inclusive, and state whether 5 and 15 are valid.

Question 2

Explain why `age>=12 || age<=18` accepts 100, correct it, and give outputs for 11,12,18,19.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first OR accepts 61 because only one side needs to be true.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

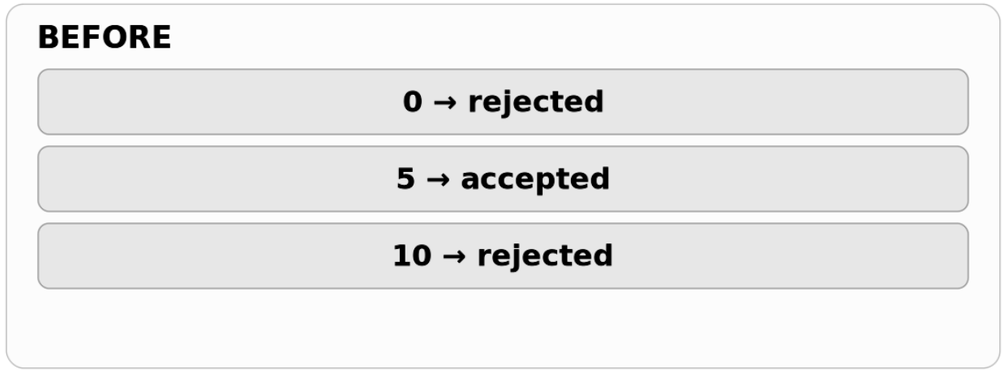
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

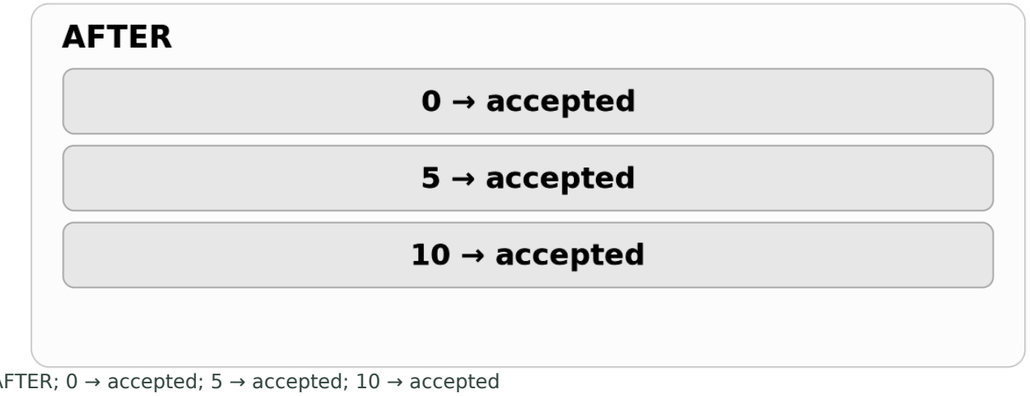
The valid endpoints were incorrectly rejected.



BEFORE; 0 → rejected; 5 → accepted; 10 → rejected

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

The valid endpoints were incorrectly rejected.



AFTER; 0 → accepted; 5 → accepted; 10 → accepted

WE DO — GUIDED PRACTICE

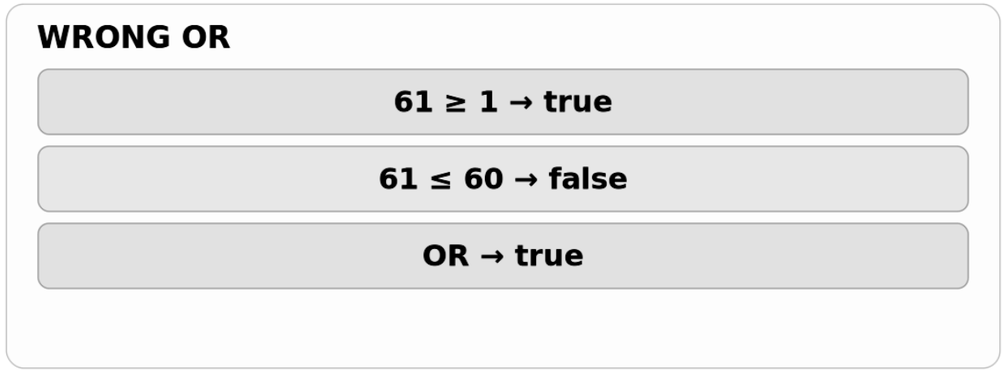
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A duration validator for 1–60 accepts 61 because it uses OR. Correct it.

OR accepts 61 because only one side needs to be true.



WRONG OR; 61 ≥ 1 → true; 61 ≤ 60 → false; OR → true